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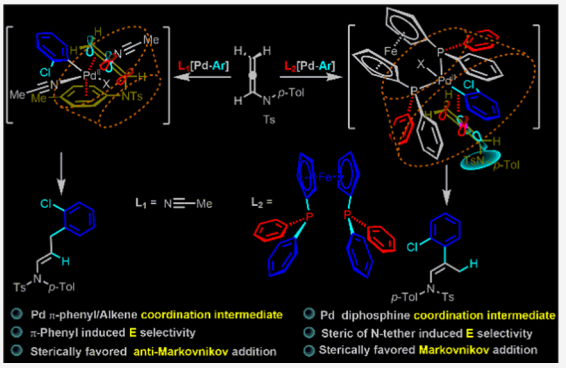
Ligand-Regulated Palladium-Catalyzed Regiodivergent Hydroarylation of the Distal Double Bond of Allenamides with Aryl Boronic Acid

Xin Du,† Huan Zhao,† Xinling Li, Lizhi Zhang, Yunhui Dong, Ping Wang, Daopeng Zhang, Qing Liu, and Hui Liu*

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ABSTRACT: The ligand-regulated regiodivergent hydroarylation of the distal double bond of allenamides with aryl boronic acid was achieved in the presence of palladium(II) catalysts, delivering a variety of functionalized enamide with excellent E selectivity and Markovnikov/anti-Markovnikov selectivity. Two possible coordination intermediates were proposed to be responsible for the regiodivergent hydroarylation: (1) The coordination **Intermediate I**, which was proposed to be formed through the coordination of MeCN, distal double bond, phenyl to palladium, led to the aryl group away from the **Intermediate I**, inducing excellent E selectivity and anti-Markovnikov selectivity. (2) A switch of regioselectivity to 1,2-**Markovnikov** hydroarylation was obtained using bidentate phosphine ligand (dppf or Xantphos). The formed coordination **Intermediate II** led to the N-tether away from the **Intermediate II** and at the trans position of aryl, resulting in excellent E selectivity and Markovnikov selectivity. Meanwhile, tentative investigation on the mechanism proved that the hydron source of this hydroarylation is more likely to be boronic acid. The transmetallation between aryl boronic acid and palladium catalyst was the initial step of this transformation.



INTRODUCTION

Regioselective processes to form one major isomer out of the possible outcomes are sought after in chemistry.[1,2] It is more attractive to develop regiodivergent reactions to achieve different regioisomers just by appropriately modulating the reaction conditions.[3] Notably, regiodivergent carbon−carbon bonds construction is of central importance to form the desired skeletal backbones.[4]

Allenamides, inducing delocalization of the lone pair electrons on nitrogen and therefore exhibiting improved stability, are ubiquitous in organic chemistry.[5] Considering the particular skeleton of allenamide bearing two C−C π bonds, controlling the site selectivity still constituted a daunting scenario, reinforcing a change in strategy.[6] On the basis of our interest in developing selective functionalization of allenamides,[7] we envision that transition-metal-catalyzed hydroarylation with high distal double-bond selectivity could deliver versatile functionalized enamides, which have emerged as increasingly valuable functional groups and highly useful building blocks in synthetic chemistry.[8] As predicted, four isomers might be alternatively products, involving Z/E and anti-Markovnikov/Markovnikov selectivity (Figure 1). Therefore, the precise control of the regioselectivity and diastereoselectivity issue remains a significant challenge.

Although a large number of hydroarylation of allenamides have been described to synthesize substituted alkenes,[9] highly selective distal hydroarylation to construct enamides is relatively underdeveloped. In 2007, Fujii and Ohno developed an Au-catalyzed intramolecular 1,2-anti-Markovnikov hydroarylation of allenamides to synthesize dihydroquinolines, which exhibited a high 6-endo selectivity due to the preferable stability of six-membered cyclization.[10]

In 2010, Kimber demonstrated intermolecular gold-catalyzed 1,2-anti-Markovnikov hydroarylation of activated allenamides, during which electron-rich aromatics and heteroaromatics are necessary (Eq. 1, Scheme 1A).[11] Very recently, Taillefer and Monnier et. al realized a similar methodology using an additive-free copper catalytic system.[12] Bandini group developed a Ni-catalyzed stereoselective synthesis of (E)-trisubstituted tertiary enamides via 1,2-Markovnikov hydroarylation of allenamides using aryl boronic

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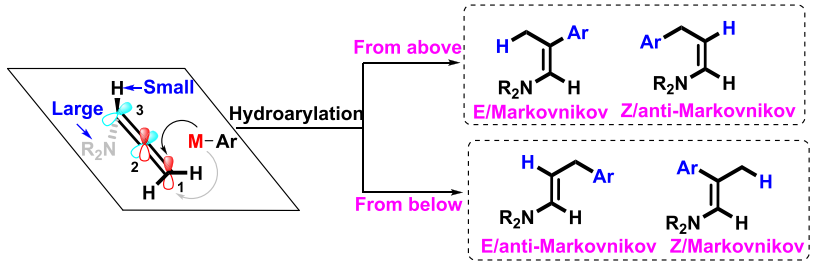
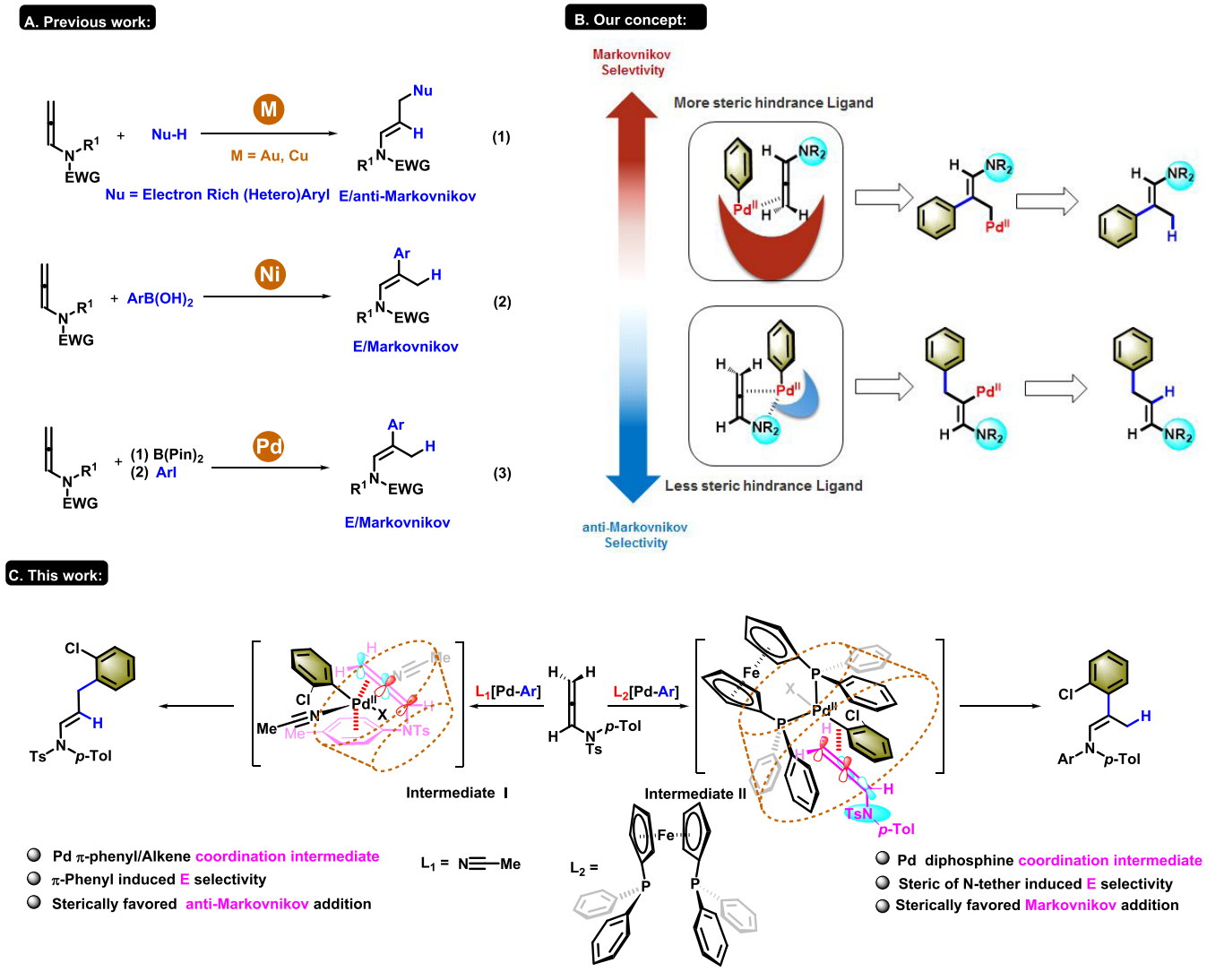


Figure 1. Hydroarylation of allenamides.

Scheme 1. Transition-Metal-Catalyzed Intermolecular Hydroarylation of Allenamides



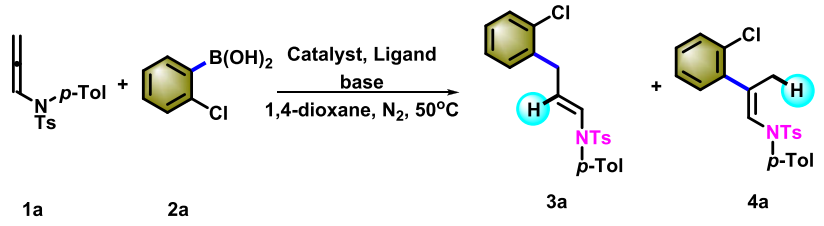
acid (Eq. 2, Scheme 1A).[13] In 2019, our group developed a palladium-catalyzed regioselective hydroarylation reaction relying on a borylation arylation strategy, and the 1,2-Markovnikov hydroarylation could be achieved via a sequential procedure by adjusting the addition order of the catalyst and iodobenzene derivatives (Eq. 3, Scheme 1A).[14] In all of these pioneering examples, excellent regioselectivities were achieved; however, only one regioisomer could be obtained, and no regiodivergent enantioselective strategy has previously been reported.

Driven by our special interest in transition-metal-catalyzed selective functionalization of allenamides, we thus envisioned whether we could devlop a ligand-regulated palladium-catalyzed hydroarylation of allenamides with high regio- and stereoselectivities (Scheme 1B). Considering the Pd−phenyl complex generated from transmetalation of Pd(II) with aryl boronic acid, a coordination intermediate might be formed via the coordination of Pd with less steric hindrance ligand, distal alkene, and N-tether, which could induce E and anti-Markovnikov selectivity (Scheme 1B, bottom).

On the other hand, another coordination intermediate might be formed through the coordination of a large ligand with palladium, which will keep the N-tether away from the more steric hindrance ligand and phenyl group, leading to **E** and

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Table 1. Optimization of Regioselective Allenamides Hydroarylations[a,b]

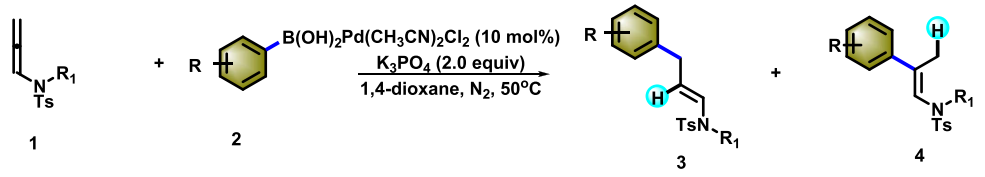


1a 2a 3a 4a

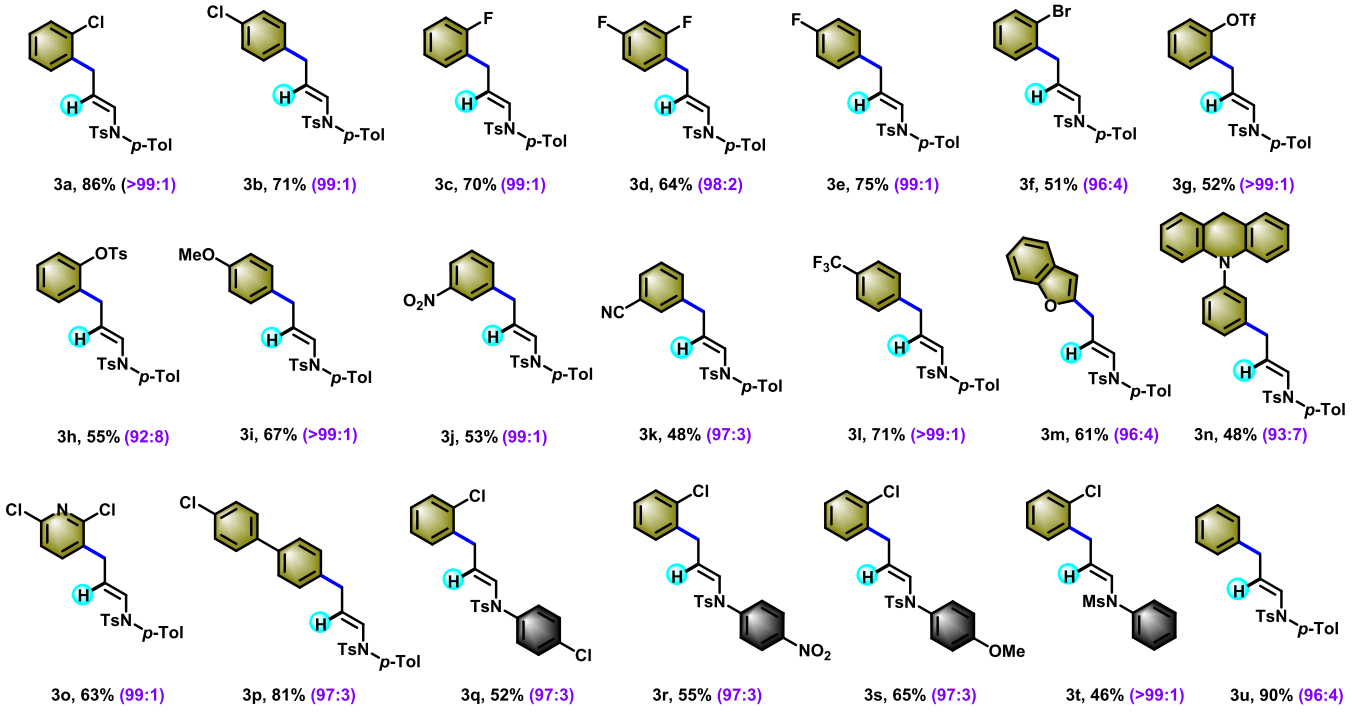
entry	catalyst	ligand	base	yield[a]	ratio of 3a:4a
1	$PdCl_2$		K_3PO_4	trace	
2	$PdCl_2$	MeCN	K_3PO_4	ND	
3	$Pd(OAc)_2$		K_3PO_4	49%	49:51
4	$Pd(OAc)_2$	2,2-Dipyridyl	K_3PO_4	45%	95:5
5	***$Pd(CH_3CN)_2Cl_2$***		***K_3PO_4***	**86%**	**99:1**
6	$Pd(CH_3CN)_2Cl_2$		Cs_2CO_3	74%	99:1
7	$Pd(CH_3CN)_2Cl_2$		DIPEA	62%	99:1
8	$Pd(CH_3CN)_2Cl_2$		CsF	54%	99:1
9	$Pd(CH_3CN)_2Cl_2$		K_2CO_3	12%	99:1
10	$Pd(OAc)_2$	DPPM	K_3PO_4	50%	84:16
11	$Pd(OAc)_2$	DPPP	K_3PO_4	48%	69:31
12	$Pd(OAc)_2$	DPPB	K_3PO_4	16%	51:49
13	***$Pd(OAc)_2$***	***Xantphos***	**K_3PO_4**	**81%**	**15:85**
14	***$Pd(OAc)_2$***	***DPPF***	**K_3PO_4**	**84%**	**14:86**

[a]Isolated yields. Reaction conditions: **1** (0.2 mmol), **2** (0.24 mmol), catalyst (10 mol %), ligand (10 mol %), base (2.0 equiv), solvent (2 mL), N_2, 50 °C. [b]The ratio of **3a:4a** in parentheses was determined by ^{1}H NMR spectroscopy.

Table 2. Substrate Scope of the Anti-Markovnikov Hydroarylation[a,b]



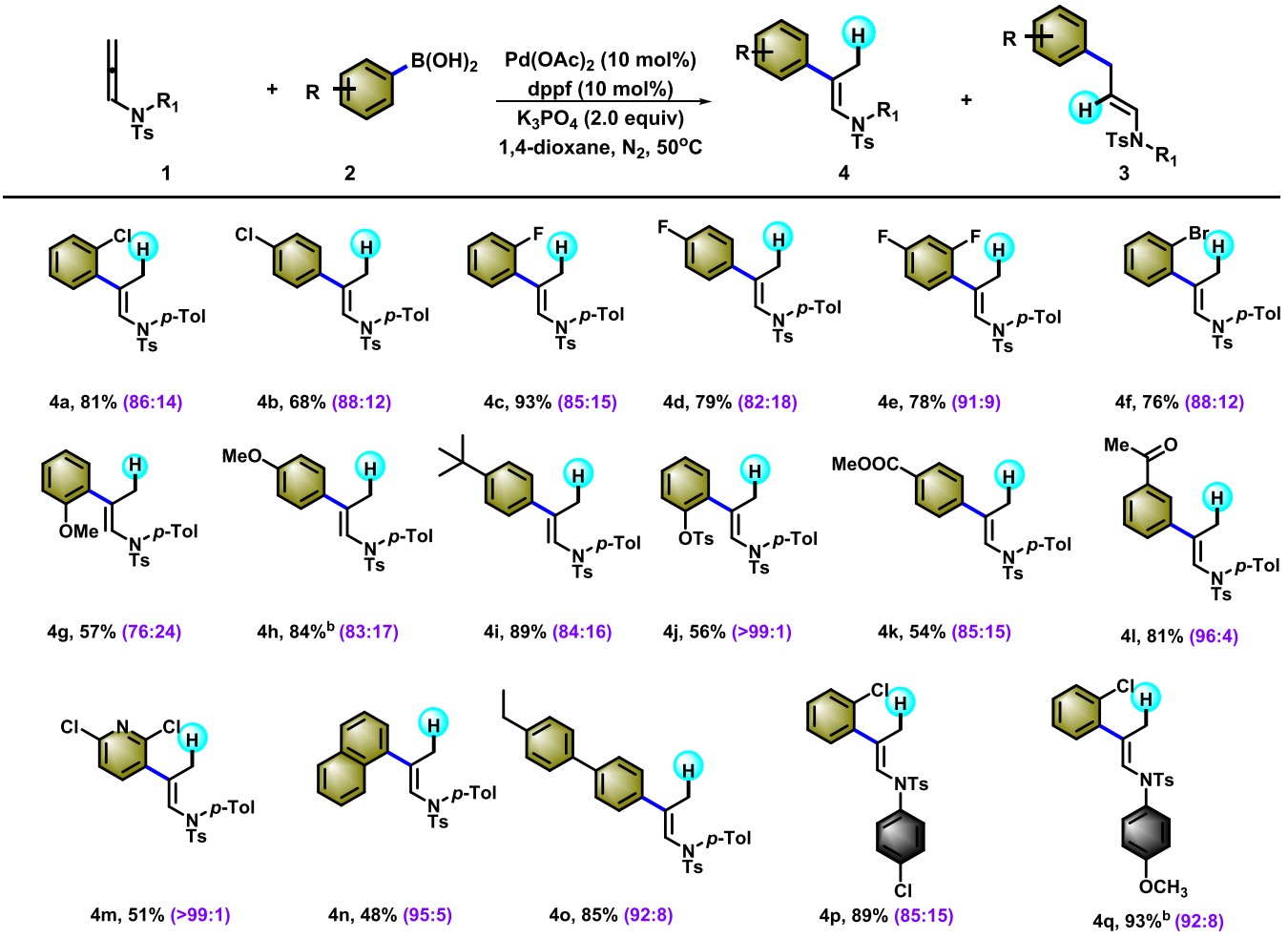
1 2 3 4



[a]Isolated yields. Standard conditions: **1** (0.2 mmol), **2** (0.24 mmol), $Pd(CH_3CN)_2Cl_2$ (10 mol %), K_3PO_4 (2.0 equiv), 1,4-dioxane (2 mL), N_2, 50 °C. [b]The ratio of **3:4** in parentheses was determined by ^{1}H NMR spectroscopy.

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Table 3. Substrate Scope of the Markovnikov Hydroarylation[a,b,c]



[a]Isolated yields. Reaction conditions: **1** (0.2 mmol), **2** (0.24 mmol), $Pd(OAc)_2$ (10 mol %), Dppf (10 mol %), K_3PO_4 (2.0 equiv), 1,4-dioxane (2 mL), N_2, 50 °C. [b]$Pd(OAc)_2$ (10 mol %), Xantphos (10 mol %). [c]The ratio of **4**:**3** in parentheses was determined by ^{1}**H NMR** spectroscopy.

Markovnikov selectivity (Scheme 1B, top). In conformity with this idea, we herein wish to report our new achievements on palladium-catalyzed regiodivergent hydroarylation of allenamides with aryl boronic acid regulated through the coordination intermediates (Scheme 1C). The **Intermediate I** was proposed to be formed through the coordination of MeCN, distal double bond, aryl group to palladium, and the internal steric hindrance of **Intermediate I** caused the aryl group far away from the coordination complex, inducing excellent **E** selectivity and **anti-Markovnikov** selectivity (Scheme 1C, left).

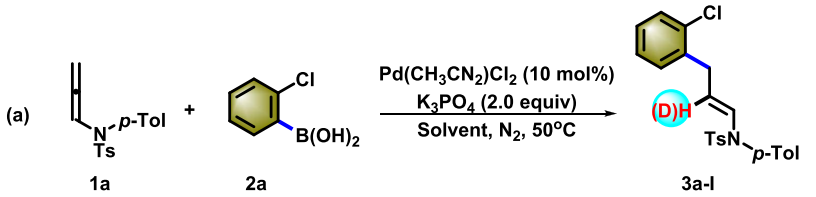
Then, **Intermediate II** was formed through coordination of dppf with Pd-aryl, resulting in the N-tether far away from the coordination complex and staying at the trans position of the aryl group, leading to excellent E selectivity and Markovnikov selectivity (Scheme 1C, right).

RESULTS AND DISCUSSION

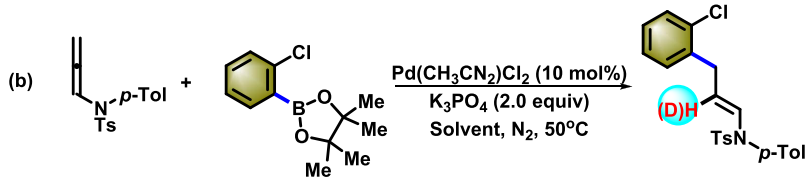
To verify our hypothesis, we tested the hydroarylation using allenamide **1a** and *o*-chloro phenylboronic acid **2a** as the model substrates. (Table 1). First of all, considering less steric hindrance complex, $PdCl_2$ without additional ligand was utilized as precatalyst. However, a trace of the desired product was detected via GCMS (entry 1). We have tested the reaction with $PdCl_2$ (10 mol %) and 3 equivalents of CH_3CN. Unfortunately, the desired product was not detected and no reaction was performed. When $Pd(OAc)_2$ was used combined with K_3PO_4 in dioxane, the hydroarylation products were isolated in a 49% yield. Although the ratio of anti-Markov addition product **3a** to Markov addition product **4a** is 49:51, excellent E selectivity was obtained (entry 3). Notably, the utilizing of ligand 2,2-dipyridyl could improve the ratio of **3a:4a** to 95:5 with a 45% yield (entry 4). To our delight, the application of $Pd(CH_3CN)_2Cl_2$ could provide the desired anti-Markov addition product **3a** in 86% yield with 99:1 selectivity (**3a:4a**). Several bases were also investigated, such as Cs_2CO_3, CsF, K_2CO_3, and DIPEA; however, no better results were obtained (see the SI for more details). Furthermore, bidentate phosphine ligand was investigated to regulate the coordination via steric hindrance to produce Markovnikov product **4a**. The use of diphosphine ligands such as dppm, dppp, and dppb provided **3a** in medium yields. Importantly, both Xantphos and dppf provided the desired **4a** in good isolated yields and Markovnikov hydroarylation selectivity. Notably, the rigid structure of dppf might be more favoring the stability of coordination **intermediate II** than the other complex leading to the good selectivity to **4a**. Thus, $Pd(CH_3CN)_2Cl_2/K_3PO_4$ and $Pd(OAc)_2$/DPPF or Xantphos/K_3PO_4 were chosen to the

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Scheme 2. Hydrogen Source Exploration Experiment



Entry	Solvent/additive	Yield (%)	deuterated ratio (%)
1	1,4-dioxane-D8	81	-
2	1,4-dioxane/D_2O	47	<5%



Entry	Solvent/additive	Yield (%)	deuterated ratio (%)
1	1,4-dioxane/D_2O	8	<5%
2	1,4-dioxane	Trace	-

optimized conditions for E selectivity and Markovnikov/anti-Markovnikov hydroarylation, respectively.

With the optimized conditions in hand, we first investigated the substrate scope of anti-Markovnikov hydroarylation (Table 2). The aryl boronic acid-bearing halides (such as F, Cl, and Br) worked smoothly to afford the multisubstituted enamide in good yields with excellent anti-Markovnikov hydroarylation selectivity (**3a−3f**). Both electron-donating (OTf, OTs, and OMe) and electron-withdrawing groups (CN, NO_2, and CF_3) at the *o*, *m*, *p*-positions of aryl boronic acid could deliver the desired products efficiently (**3g−3l**).

Remarkably, the active Br, OTs, and OTf, which could be further transformed to more valuable molecules via coupling reactions, could survive very well under the standard conditions.

Furthermore, heteroaryl boronic acids (such as bearing benzofuran, 2,6-di-Cl-pyridine, and 10-phenyl-9,10-dihydroacridine) were also examined and the corresponding products were obtained in moderate yields with good anti-Markovnikov hydroarylation selectivity (**3m−3o**). The reaction of (4′-chloro-[1,1′-biphenyl]-4-yl)boronic acid with the allenamide afforded the enamide derivate in 81% yield with 97:3 selectivity (**3p**). Furthermore, we also evaluated the substitution of R_1 on the allenamides. The results showed that when R_1 was aryl group with no matter electron-neutral, -rich, or -poor substitutions, the transformation of allenamides affords the corresponding anti-Markovnikov addition products in moderate yields (**3q−3s**). Besides Ts-protected allenamide, Ms was also well suitable for the anti-Markovnikov addition transformation albeit with a relatively lower yield than Ts derivates (**3t**). In addition, we also used phenylboronic acid for the substrate and obtained a confused result as shown below (**3u**). Xantphos leads C_1 selectivity, but dppf leads to C_2 selectivity, and the reason is not clear at present.

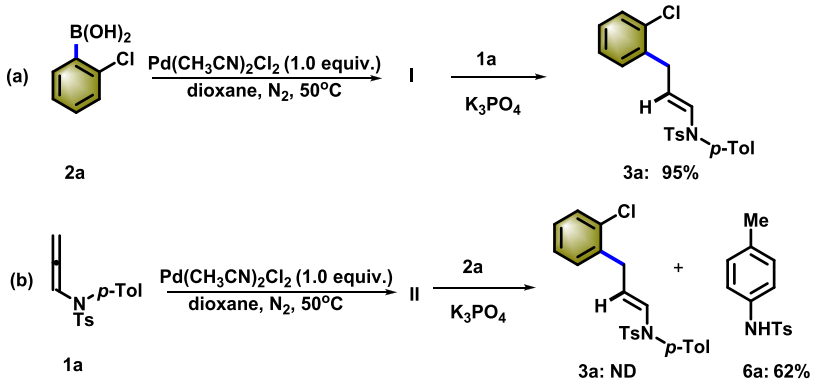
Subsequently, we moved to explore the substrate scope of Markovnikov hydroarylation (Table 3). First of all, a variety of halides are well tolerated, including fluoro, chloro, and bromo (**4a−4f**). Gratifyingly, aryl boronic acids with both electron-donating (**4g−4j**) and electron-withdrawing (**4k−4l**) groups were examined and the transformation proceeded smoothly to deliver the corresponding Markovnikov products in moderate to high yields, with good regioselectivity.

In particular, the active group in coupling reactions such as OTs and Bromo, survived well in this strategy. Furthermore, pyridyl (**4m**)-, naphthyl (**4n**)-, and biphenyl (**4o**)-substituted boronic acid successfully afforded the desired Markovnikov hydroarylation products in good yields with excellent selectivity. Two allenamides were also investigated, which showed good compatibility to this strategy (**4p−4q**).

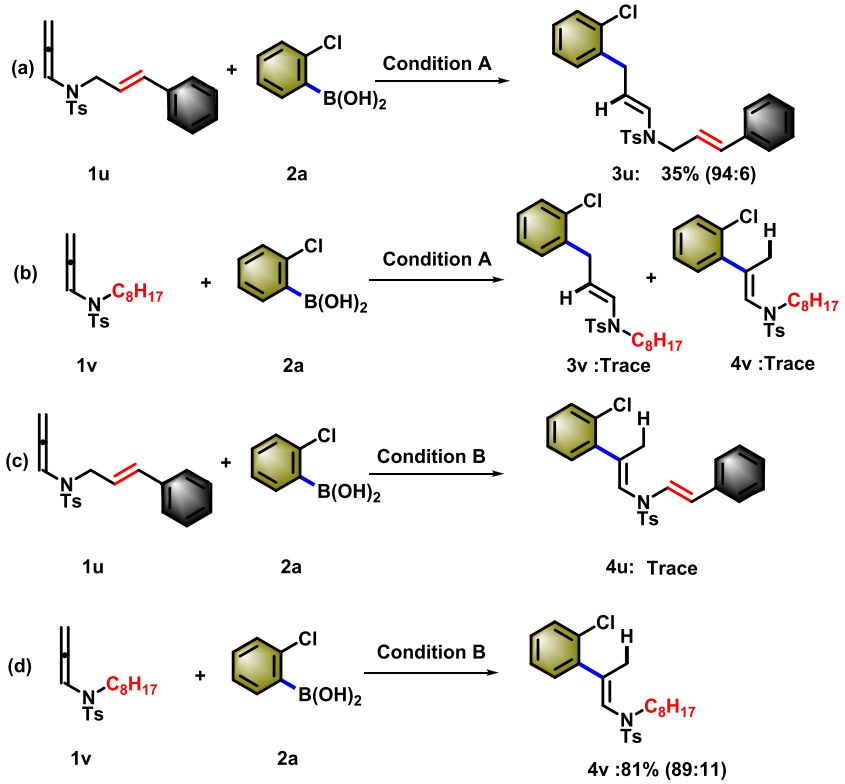
To get insight into the reaction mechanism, a series of preliminary mechanistic exploration experiments were performed. First, some controlled experiments were conducted to investigate the source of hydrogen in this transformation (Scheme 2). When the reaction was carried out in 1,4-dioxane-D8, the corresponding anti-Markov addition product **3a−I** was obtained in an 81% yield without deuterium labeling, indicating that the hydrogen did not originate from solvent (Scheme 2a, entry 1). However, when the reaction was conducted in 1,4-dioxane (2 mL) with D_2O (2.0 equiv), product **3a−I** was isolated in a 47% yield with quite less deuterium incorporation (Scheme 2a, entry 2). This observation made us speculate that the source of hydrogen is not from the residual H_2O of solvent but maybe from boronic acid. Subsequently, 2-(2-chlorophenyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane **5a** was synthesized and subjected to the mixed solvent of 1,4-dioxane (2 mL) and D_2O (2.0 equiv) under standard condition. Notably, the reaction afforded compound **3a−I** in 8% yield with less than 5% deuterium incorporation. (Scheme 2b, entry 1). Nevertheless, when the reaction of **1a** and **5a** was conducted under the standard condition, the transformation could not be proceeded (Scheme 2b, entry 2). On the basis of these results, the hydrogen for

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Scheme 3. Initial-Step Investigations



Scheme 4. Investigation of the Coordination Effect of the Phenyl Group in intermediate I[a,b]



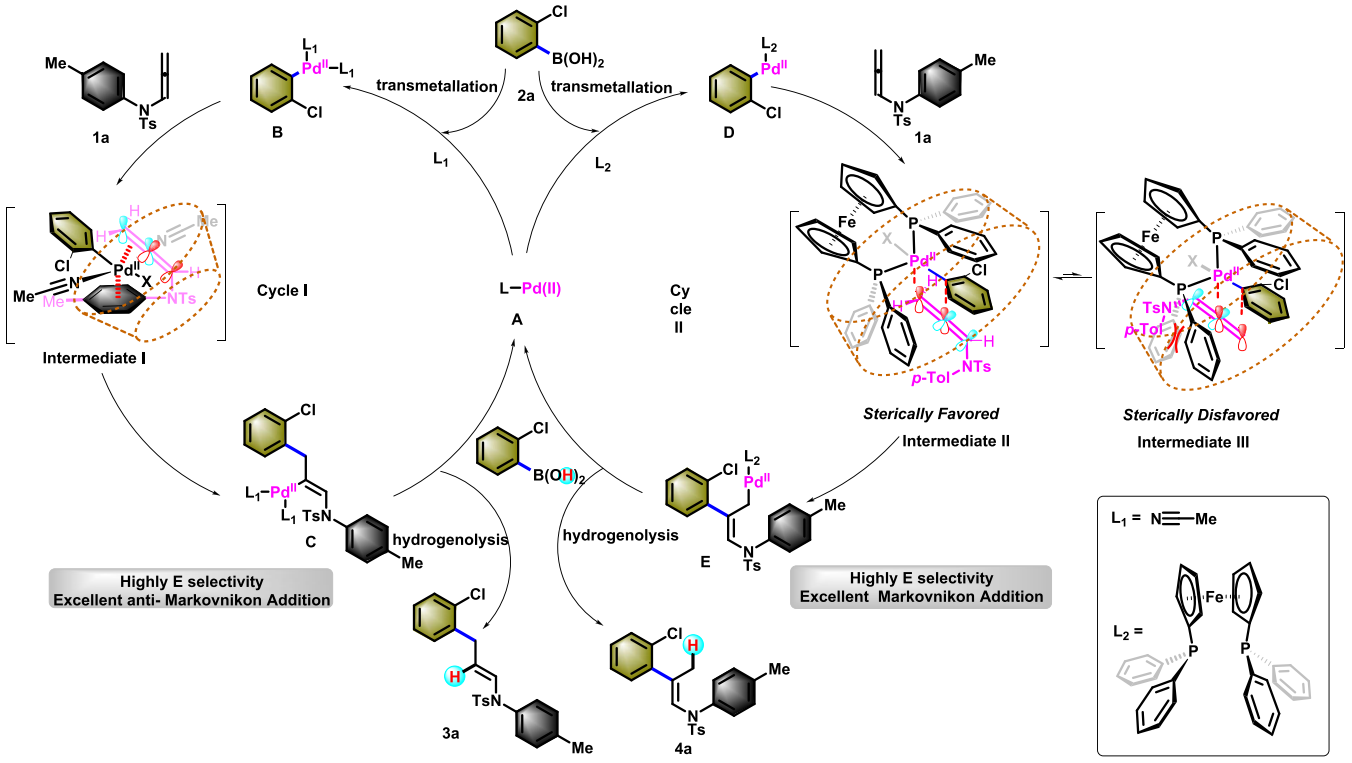
[a]Condition A: **1u**/**1v** (0.2 mmol), **2a** (0.24 mmol), $Pd(CH_3CN)_2Cl_2$ (10 mol %), K_3PO_4 (2.0 equiv), 1,4-dioxane (2 mL), N_2, 50 °C. [b]Condition B: **1u**/**1v** (0.2 mmol), **2a** (0.24 mmol), $Pd(OAc)_2$ (10 mol %), Dppf (10 mol %), K_3PO_4 (2.0 equiv), 1,4-dioxane (2 mL), N_2, 50 °C.

hydroarylation was considered to be probably from aryl boronic acid. The residual H_2O of solvent could also play as the hydrogen source; however, the yields decreased sharply owing to the decomposition of allenamide in the presence of water.

In addition, to investigate the initial step of this transformation, several control experiments were carried out. As shown in Scheme 3, *o*-chlorophenylboronic acid **2a** was added to 1.0 equiv. $PdCl_2(MeCN)_2$ in dioxane and stirred for 0.5 h; then, the allenamide **1a** and K_3PO_4 were added. After stirring for 1 h, the anti-Markovnikov product **3a** was obtained in a 95% yield (Scheme 3a). On the other hand, allenamide **1a** was subjected to 1.0 equiv. $PdCl_2(MeCN)_2$ for 0.5 h before the addition of *o*-chloro phenyl acid **2a** and K_3PO_4. However, no hydroarylation product was generated, but the decomposed product 4-methyl-*N*-(*p*-tolyl)benzenesulfonamide was isolated in 62% (Scheme 3b). These results indicated that the initial step of this transmetalation is the transmetallation between palladium(II) and aryl boronic acid. To investigate the proposed palladium complex **intermediate I**, substrates **1u** and **1v** were synthesized and subjected to the standard conditions (Scheme 4a–d). The substrate **1u** bearing a 3-phenylprop-2-ene group (instead of phenyl) could undergo anti-Markovnikov hydroarylation to deliver the desired product **3u** in 35% yield with 94:6 selectivity; however, when the phenyl tether of allenamides was replaced with octanyl, the reaction of **1v** could not afford the hydroarylation products efficiently and only a trace of the desired hydroarylation product was detected via GCMS. We also have tried hydroarylation of **1v** under $Pd(OAc)_2$ and dppf condition, and the corresponding Markov addition product **4v** was obtained in an 81% yield with 89:11 Markovnikov selectivity. However,

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Scheme 5. Possible Mechanism



the reaction of **1u** did not perform well, and we could not isolate an obvious product. Both of these results indicated that the coordinating effect of the phenyl group was the key to the anti-Markovnikov selectivity. This result well coincided with our proposed coordination **intermediate I** in Scheme 1C.

Based on the results above, a possible mechanism for this ligand-controlled palladium-catalyzed regiodivergent hydroarylation of allenamides with aryl boronic acid was proposed as depicted in Scheme 5. In cycle **I**, transmetalation between Pd(II) catalyst and *o*-chlorophenylboronic acid **2a** generated the Pd−phenyl complex **B**, which then coordinated to allenamide **1a** to form the coordination **intermediate I** with MeCN, distal double bond, phenyl of allenamide. The specially coordinated structure of **intermediate I** induced excellent E selectivity and anti-Markovnikov addition selectivity of hydroarylation to deliver intermediate **C**. The following hydrogenolysis of palladium complex **C** with the hydrogen from the aryl boronic acid afforded the anti-Markovnikov product **3a** with regeneration of L_1-Pd(II) catalyst. On the other hand, when the large ligand dppf was used, the transmetallation could generate palladium(II) species **D**. The coordination of dppf with complex **D** formed a complex in which the N-tether was pushed out of the complex and at the trans position to aryl group, resulting in excellent E selectivity and Markovnikov selectivity as depicted in coordination **intermediate II**. Meanwhile, **intermediate III** is sterically disfavored than **intermediate II** as shown in cycle **II** of Scheme 4. Subsequently, the coordination **intermediate II** underwent Markovnikov addition to afford the Pd(II) intermediate **E**. Finally, hydrogenolysis of intermediate **E** gave the desired product **4a** with regeneration of the L_2-Pd(II) catalyst.

CONCLUSIONS

Overall, we have developed the palladium-catalyzed regiodivergent hydroarylation of allenamides through ligand regulation. Various allenamides reacted with aryl boronic acids to produce the corresponding **Markovnikov** or **anti-Markovnikov** hydroarylation products (functionalized enamide) in high yields with excellent regioselectivity. Through ligand regulation, the hydroarylation regioselectivity of the distal double bond of allenamide is controllable.[15] Controlled experiments on the mechanism indicated that the hydrogen source may come from boronic acid for the first time. Two possible coordination intermediates were proposed to be responsible for the excellent E selectivity and Markovnikov/anti-Markovnikov selectivity: (1) The coordination **intermediate I**, which was proposed to be formed through the coordination of MeCN, distal double bond, phenyl to palladium, led to the aryl group away from **intermediate I**, inducing excellent E selectivity and anti-Markovnikov selectivity. (2) A switch of regioselectivity to 1,2-**Markovnikov** hydroarylation was obtained using bidentate phosphine ligand (dppf or Xantphos). The formed coordination **intermediate II** led to the N-tether away from **intermediate II** and at the trans position of aryl, resulting in excellent E selectivity and Markovnikov selectivity. Further investigation on the mechanism of regiodivergent hydroarylation and application of the enamides will be in due course.

EXPERIMENTAL SECTION

General Information. Commercially available reagents were used without further purification. Purifications of reactions products were carried out by flash chromatography using Merck silica gel (40−63 μm). ^{1}H NMR (400 MHz) and ^{13}C NMR (100 MHz) spectra were recorded on a Bruker Avance 400 MHz spectrometer. Chemical shifts are reported in parts per million (ppm, δ) downfield from residual solvents peaks, and coupling constants are reported in hertz (Hz).

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Splitting patterns are designated as singlet (s), doublet (d), triplet (t), etc. Splitting patterns that could not be interpreted or easily visualized are designated as multiplet (m). Electrospray mass spectra were obtained using an ESI/TOF-Q Mariner mass spectrometer. Unless otherwise noted, all other commercially available reagents and solvents were used without further purification. For reactions that require heating, the heat source is an oil bath.

Synthesis. *4-Methyl-N-(2l5-propa-1,2-dien-1-yl)-N-(p-tolyl)-benzenesulfonamide (**1a**).* To a solution of p-toluidine (3.2 g, 30 mmol) in DCM (15 mL) at room temperature was added TsCl (6.3 g, 33 mmol). Then, a solution of triethylamine (9.1 g, 90 mmol) in 15 mL of DCM was added dropwise through a constant-pressure funnel under vigorous stirring. The mixture was stirred at room temperature overnight and then washed three times with water. The organic phase was combined and dried over Na_2SO_4 to afford 4-methyl-*N*-(*p*-tolyl)benzenesulfonamide (7.1 g, 27 mmol) in a 91% yield as a white solid. The 4-methyl-*N*-(*p*-tolyl)benzenesulfonamide (1.83 g, 7 mmol) was dissolved in 40 mL of MeCN, and 3-bromoprop-1-yne (0.7 mL, 9.1 mmol), potassium carbonate (1.94 g, 14 mmol), and potassium iodide (0.58 g, 3.5 mmol) were added successively. The mixture was stirred for 2 h and refluxed at 60 °C. Then, the reaction system was quenched with water, and the aqueous layer was extracted three times with EtOAc. Drying over Na_2SO_4 and concentration in vacuo afforded the crude material, which was then purified by silica gel column chromatography (PE/EA = 5:1) to give 4-methyl-*N*-(prop-2-yn-1-yl)-*N*-(*p*-tolyl)benzenesulfonamide (1.86 g, 6.21 mmol, 89% yield) as a white solid. The solution of tBuOK (0.67 g, 6 mmol, 1.5 equiv) was stirred in dry THF (10 mL) at −5 °C under a nitrogen atmosphere. The 4-methyl-*N*-(prop-2-yn-1-yl)-*N*-(*p*-tolyl)benzenesulfonamide (1.2 g, 4 mmol) was dissolved in 5 mL of THF and added dropwise into the solution over a period of 30 min. The organic phase was combined, dried over Na_2SO_4, evaporated, and purified by flash chromatography (PE/EA = 10:1) to give compound **1a** (0.93 g, 3.1 mmol, 78% yield). The following allene derivatives were prepared from the corresponding amine compounds following procedure A. The allene derivative **1q**−**1v** was prepared from previously reported methods.[16]

*2-(4,4,5,5-Tetramethyl-1,3,2-dioxaborolan-2-yl)phenyl Trifluoromethanesulfonate (**1g**).* To a solution of (2-hydroxyphenyl)boronic acid (1.10 g, 8.0 mmol) in dichloromethane (15 mL) was added pinacol (1.04 g, 8.8 mmol) at room temperature. After stirring the mixture for 2 h at the same temperature, water (ca. 10 mL) was added and then extracted with dichloromethane (ca. 30 mL × 3). The combined organic extract was washed with brine (ca. 30 mL × 1), dried over Na_2SO_4, and, after filtration, the filtrate was concentrated under reduced pressure. The residual colorless oil containing 2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenol was used in the next reaction without further purification. Chromatographically purified or commercially available 2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenol could be equally used for further derivatization. To a solution of crude 2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phenol in DCM (25 mL) were added *N*,*N*-diisopropylethylamine (1.57 mL, 9 mmol) and trifluoromethanesulfonic anhydride (1.21 mL, 7.2 mmol) at −78 °C. After stirring the mixture for 3 h at the same temperature, water (ca. 10 mL) was added and then extracted with dichloromethane (ca. 10 mL × 3). The combined organic extract was dried over Na_2SO_4 and, after filtration, the filtrate was concentrated under reduced pressure. The residual oil was purified by passing through a short pad of silica gel (*n*-hexane/ethyl acetate = 10/1) followed by Kugelrohr distillation (ca. 150 °C/2.0 torr) to give **1g** (1.82 g, 5.17 mmol, 86% from (2-hydroxyphenyl)boronic acid) as a colorless oil. The precursors of compound **1h** were prepared following previously reported methods.[17]

General Procedures for Products 3. Allenamide (0.20 mmol), phenylboric acid (0.24 mmol), $Pd(CH_3CN)_2Cl_2$ (5.0 mg, 0.02 mmol), and K_3PO_4 (85.0 mg, 0. 4 mmol) were added to a reaction tube, Then, toluene 1,4-dioxane (2 mL) was added under nitrogen atmosphere. The resulting mixture was stirred at 50 °C for 1 h (the heat source is an oil bath). After cooling the reaction mixture at room temperature, the aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic phase was dried over Na_2SO_4 and then concentrated in vacuo. The mixture was purified by silica gel column chromatography to give the products.

*(E)-N-(3-(2-Chlorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**3a**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3a** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (70.9 mg, 86%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.47 (d, J = 7.6 Hz, 2H), 7.24-7.18 (m, 3H), 7.08-7.01 (m, 5H), 6.96 (d, J = 14 Hz, 1H), 6.75 (d, J = 7.6 Hz, 2H), 4.44-4.37 (m, 1H), 3.27 (d, J = 6.8 Hz, 2H), 2.36 (s, 3H), 2.26 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, CDCl3): δ = 143.8, 139.0, 138.1, 135.9, 133.9, 133.8, 130.8, 130.1, 130.1, 129.8, 129.6, 129.4, 127.6, 127.5, 126.8, 108.9, 33.7, 21.7, 21.3. HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{23}H_{23}ClNO_2S$, 412.1138; found 412.1138.

*(E)-N-(3-(4-Chlorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**3b**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3b** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (58.5 mg, 71%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.55 (d, J = 7.6 Hz, 2H), 7.28 (d, J = 7.6 Hz, 2H), 7.20 (d, J = 7.2 Hz, 2H), 7.12 (d, J = 7.6 Hz, 2H), 7.04 (d, J = 14 Hz, 1H), 6.99 (d, J = 7.6 Hz, 2H), 6.83 (d, J = 7.2 Hz, 2H), 4.50−4.43 (m, 1H), 3.24 (d, J = 7.2 Hz, 2H), 2.45 (s, 3H), 2.33 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 143.8, 139.1, 139.0, 135.9, 133.8, 131.8, 130.4, 130.2,129.8, 129.6, 129.6, 128.4, 127.5, 11 0.2, 35.4, 21.7, 21.3. HRMS (ESI/OTOF-Q) m/z: [M + Na]$^+$ calcd for $C_{23}H_{22}ClNO_2SNa$, 434.0957; found 434.0954.

*(E)-N-(3-(2-Fluorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**3c**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3c** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (55.4 mg, 70%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.47 (d, J = 7.6 Hz, 2H), 7.19 (d, J = 8.0 Hz, 2H), 7.10−7.02 (m, 3H), 6.99 (d, J = 4.4 Hz, 1H), 6.95 (d, J = 7.2 Hz, 2H), 6.92−6.87 (m, 1H), 6.75 (d, J = 7.6 Hz, 2H), 4.45−4.38 (m, 1H), 3.22 (d, J = 6.8 Hz, 2H), 2.36 (s, 3H), 2.25 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 160.9 (d, J = 243.9 Hz), 143.7, 138.9, 136.0, 134.0, 130.5, 130.3 (d, J = 4.7 Hz), 130.1, 129.8, 129.5, 127.8 (d, J = 10 Hz), 127.5, 124.0 (d, J = 3.5 Hz), 115.1 (d, J = 21.8 Hz), 109.3, 29.0 (d, J = 3.3 Hz), 21.6, 21.2. ^{19}F NMR (376 MHz, $CDCl_3$): δ = −118.6. HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{23}H_{23}FNO_2S$, 396.1434; found 396.1434.

*(E)-N-(3-(2,4-Difluorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**3d**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3d** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (52.9 mg, 64%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.54 (d, J = 7.6 Hz, 2H), 7.27 (d, J = 8 Hz, 2H), 7.12 (d, J = 8 Hz, 2H), 7.06−6.98 (m, 2H), 6.82 (d, J = 7.6 Hz, 2H), 6.78−6.70 (m, 2H), 4.48−4.41 (m, 1H), 3.25 (d, J = 7.2 Hz, 2H), 2.44 (s, 3H), 2.33 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 161.5 (dd, J = 236.3, 11.8 Hz), 160.6 (dd, J = 234.9, 11.5 Hz), 143.8, 139.0, 135.9, 133.8, 130.7 (dd, J = 9.4, 6.3 Hz,), 130.6, 130.2, 129.8, 129.6, 127.5, 123.4 (dd, J = 15.8, 3.8 Hz), 111.0 (dd, J = 20.8, 3.8 Hz), 108.8, 103.6 (t, J = 25.5 Hz), 28.5 (d, J = 2.8 Hz), 21.6, 21.2. ^{19}F NMR (376 MHz, $CDCl_3$): δ = −113.3 (s, 1F), −114.4 (s, 1F). HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{23}H_{22}F_2NO_2S$, 414.1339; found 414.1342.

*(E)-N-(3-(4-Fluorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**3e**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3e** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (59.3 mg, 75%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.55 (d, J = 7.6 Hz, 2H), 7.28 (d, J = 8 Hz, 2H), 7.12 (d, J = 8 Hz, 2H), 7.05-7.01 (m, 3H), 6.93 (d, J = 8.8 Hz, 1H), 6.91 (d, J = 8 Hz, 1H), 6.83 (d, J = 7.6 Hz, 2H), 4.52-4.44 (m, 1H), 3.25 (d, J = 7.2 Hz, 2H), 2.45 (s, 3H), 2.33 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 161.4 (d, J = 242.3 Hz), 143.8, 139.0, 136.2 (d, J = 3.1 Hz), 135.9, 133.8, 130.2, 130.1, 129.8, 129.6 (d, J = 8.1 Hz), 129.6, 127.5, 115.1(d, J = 21 Hz), 110.6, 35.2, 21.6, 21.2. ^{19}F NMR (376 MHz, $CDCl_3$): δ = −117.3.

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HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{23}H_{22}FNO_2SNa$, 418.1253; found 418.1249.

*(E)-N-(3-(2-Bromophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**3f**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3f** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (46.6 mg, 51%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.55 (d, J = 7.6 Hz, 2H), 7.49 (d, J = 7.6 Hz, 1H), 7.26 (d, J = 7.6 Hz, 2H), 7.21-7.18 (m, 1H), 7.13-7.02 (m, 5H), 6.84 (d, J = 8 Hz, 2H), 4.52−4.45 (m, 1H), 3.38 (d, J = 6.8 Hz, 2H), 2.43 (s, 3H), 2.33(s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 143.8, 139.8, 139.0, 135.9, 133.9, 132.7, 130.9, 130.2, 130.1, 130.0, 129.6, 127.8, 127.5, 127.5, 124.5, 108.9, 36.3, 21.6, 21.2. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{23}H_{23}BrNO_2S$, 456.0633; found 456.0638.

*(E)-2-(3-((4-Methyl-N-(p-tolyl)phenyl)sulfonamido)allyl)phenyl Trifluoromethanesulfonate (**3g**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3g** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (54.7 mg, 52%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.55 (d, J = 8.0 Hz, 2H), 7.27 (d, J = 8.0 Hz, 4H), 7.21-7.18 (m, 2H), 7.13 (d, J = 8 Hz, 2H), 7.08 (d, J = 14 Hz, 1H), 6.84 (d, J = 7.6 Hz, 2H), 4.47-4.39 (m, 1H), 3.38 (d, J = 6.8 Hz, 2H), 2.44 (s, 3H), 2.33 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 147.8, 143.9, 139.1, 135.8, 133.7, 133.5, 131.5, 131.1, 130.2, 129.8, 129.6, 128.4, 128.1, 127.6, 121.3, 107.5, 30.0, 21.6, 21.2. ^{19}F NMR (376 MHz, $CDCl_3$): δ = −73.9 (s, 3F). HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{24}H_{22}F_3NO_5S_2Na$, 548.0789; found 548.0792.

*(E)-2-(3-(4-Methyl-N-(p-tolyl)phenylsulfonamido)allyl)phenyl 4-Methylbenzenesulfonate (**3h**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3h** by column chromatography (PE/EA = 5:1) on silica gel as a yellow oil (60.2 mg, 55%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.71 (d, J = 8 Hz, 2H), 7.55 (d, J = 7.6 Hz, 2H), 7.32−7.26 (m, 4H), 7.14−7.10 (m, 4H), 7.04 (d, J = 7.2 Hz, 1H), 7.00−6.93 (m, 2H), 6.81 (d, J = 8 Hz, 2H), 4.39−4.32 (m, 1H), 3.11 (d, J = 7.2 Hz, 2H), 2.44 (d, J = 4 Hz, 6H), 2.32 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 147.7, 145.5, 143.8, 139.0, 135.8,134.0, 133.8, 133.0, 130.8, 130.3, 130.1, 129.9, 129.8, 129.6, 128.3, 127.5, 127.4, 127.1, 122.3, 108.7, 29.9, 21.8, 21.7, 21.2. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{30}H_{29}NO_5S_2Na$, 570.1385; found 570.1382.

*(E)-N-(3-(4-methoxyphenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**3i**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3i** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (54.6 mg, 67%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.56 (d, J = 7.6 Hz, 2H), 7.28 (d, J = 8 Hz, 2H), 7.11 (d, J = 8 Hz, 2H), 7.01 (d, J = 13.2 Hz, 1H), 7.00 (d, J = 7.2 Hz, 2H), 6.83 (d, J = 8 Hz, 2H), 6.79 (d, J = 8 Hz, 2H), 4.54−4.47 (m, 1H), 3.77 (s, 3H), 3.22 (d, J = 7.2 Hz, 2H), 2.45 (s, 3H), 2.33 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 157.9, 143.7, 139.0, 136.0, 134.0, 132.6, 130.1, 130.0,129.8, 129.7, 129.6, 129.2, 127.5, 113.7, 111.4, 55.3, 21.7, 21.2. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{24}H_{25}NO_3SNa$, 430.1453; found 430.1457.

*(E)-4-Methyl-N-(3-(3-nitrophenyl)prop-1-en-1-yl)-N-(p-tolyl)-benzenesulfonamide (**3j**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3j** by column chromatography (PE/EA = 10:1) on silica gel as a yellow solid (44.8 mg, 53%). ^1H NMR (400 MHz, $CDCl_3$): δ = 8.04 (d, J = 3.6 Hz, 1H), 7.94 (s, 1H), 7.57 (d, J = 8 Hz, 2H), 7.42 (d, J = 4.8 Hz, 2H), 7.30 (d, J = 8 Hz, 2H), 7.13 (d, J = 7.6 Hz, 2H), 7.11 (d, J = 9.2 Hz, 1H), 6.84 (d, J = 8 Hz, 2H), 4.51−4.44 (m, 1H), 3.39 (d, J = 7.2 Hz, 2H), 2.45 (s, 3H), 2.33 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 148.3, 144.0, 142.8,139.2, 135.7, 134.6, 133.6, 131.3,130.3, 129.8, 129.7, 129.3, 127.5, 123.1, 121.3, 108.7, 35.6, 21.7, 21.2. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{23}H_{22}N_2O_4SNa$, 445.1198; found 445.1201.

*(E)-N-(3-(3-Cyanophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**3k**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3k** by column chromatography (PE/EA = 10:1) on silica gel as a yellow solid (38.6 mg, 48%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.56 (d, J = 7.6 Hz, 2H), 7.46 (d, J = 7.2 Hz, 1H), 7.37−7.30 (m, 5H), 7.14 (d, J = 8 Hz, 2H), 7.08 (d, J = 14 Hz, 1H), 6.84 (d, J = 8 Hz, 2H), 4.48-4.41 (m, 1H), 3.32 (d, J = 7.2 Hz, 2H), 2.46 (s, 3H), 2.34 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 144.1, 142.2, 139.2, 135.7, 133.6, 133.0, 131.7, 131.2,130.2, 130.0, 129.8, 129.7, 129.2, 127.5, 119.0, 112.3, 108.7, 35.6, 21.7, 21.3. HRMS (ESI-TOF-Q) m/z: $[M + Na]^+$ calcd for $C_{24}H_{22}N_2O_2SNa$, 425.1300; found 425.1304.

*(E)-4-Methyl-N-(p-tolyl)-N-(3-(2-(trifluoromethyl)phenyl)prop-1-en-1-yl)benzenesulfonamide (**3l**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3l** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (63.3 mg, 71%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.56 (d, J = 8.0 Hz, 2H), 7.49 (d, J = 7.6 Hz, 2H), 7.28 (d, J = 8.0 Hz, 2H), 7.18 (d, J = 7.6 Hz, 2H), 7.12 (d, J = 8.0 Hz, 2H), 7.09 (d, J = 16.4 Hz, 1H), 6.84 (d, J = 7.6 Hz, 2H), 4.51−4.44 (m, 1H), 3.34 (d, J = 7.2 Hz, 2H), 2.45 (s, 3H), 2.33 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 144.8, 143.9, 139.1, 135.9, 133.7, 130.7, 130.2, 129.8, 129.6, 128.6, 127.5, 125.5 (q, J = 3.7 Hz), 109.4, 35.9, 21.6, 21.2. ^{19}F NMR (376 MHz, $CDCl_3$): δ = −62.3 (s,3F). HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{23}F_3NO_2S$, 446.1402; found 446.1408.

*(E)-N-(3-(Benzofuran-2-yl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**3m**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3m** by column chromatography (PE/EA = 10:1) on silica gel as a yellow solid (50.9 mg, 61%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.57 (d, J = 7.6 Hz, 2H), 7.45 (d, J = 7.6 Hz, 1H), 7.38 (d, J = 7.6 Hz, 1H), 7.27 (d, J = 8 Hz, 2H), 7.22-7.18 (m, 2H), 7.15 (d, J = 8.8 Hz, 3H), 6.86 (d, J = 7.2 Hz, 2H), 6.25 (s, 1H), 4.56−4.49 (m, 1H), 3.43 (d, J = 7.2 Hz, 2H), 2.44 (s, 3H), 2.35 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 157.7, 154.7, 143.9, 139.2, 135.9, 133.7, 131.5, 130.2, 129.9,129.6, 128.7, 127.5, 123.3, 122.5, 120.3, 110.8, 106.0, 102.3, 29.1, 21.7, 21.3. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{25}H_{24}NO_3S$, 418.1477; found 418.1476.

*(E)-N-(3-(3-(Acridin-10(9H)-yl)phenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**3n**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3n** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (53.4 mg, 48%). ^1H NMR (400 MHz, $CDCl_3$): δ = 8.14 (d, J = 8 Hz, 2H), 7.60 (d, J = 8 Hz, 2H), 7.55−7.52 (m, 1H), 7.44−7.35 (m, 7H), 7.30−7.26 (m, 8H), 7.17−7.13 (m, 3H), 6.90 (d, J = 7.6 Hz, 2H), 4.65−4.58 (m, 1H), 3.42 (d, J = 7.6 Hz, 2H), 2.44 (s, 3H), 2.36 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 143.8, 140.9, 140.1, 139.1, 136.0, 135.6, 133.9, 130.6, 130.2, 129.9, 129.7, 129.6, 127.6, 127.0, 125.9, 123.2, 120.3, 119.8, 110.1, 109.7, 35.8, 21.7, 21.3. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{36}H_{33}N_2O_2S$, 557.2263; found 557.2266.

*(E)-N-(3-(2,6-Dichloropyridin-3-yl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**3o**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3o** by column chromatography (PE/EA = 10:1) on silica gel as a yellow oil (56.4 mg, 63%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.54 (d, J = 8 Hz, 2H), 7.38 (d, J = 7.6 Hz, 1H), 7.28 (d, J = 8.4 Hz, 2H), 7.18-7.10 (m, 4H), 6.82(d, J = 8 Hz, 2H), 4.40−4.32 (m, 1H), 3.35 (d, J = 7.2 Hz, 2H), 2.44 (s, 3H), 2.34 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 149.8, 147.9, 144.0, 140.8, 139.3, 135.6, 133.8, 133.5, 131.9,130.3, 129.7, 129.6, 127.5, 123.1, 106.2, 32.7, 21.7, 21.3. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{22}H_{21}Cl_2N_2O_2S$, 447.0701; found 447.0705.

*(E)-N-(3-(4′-Chloro-[1,1′-biphenyl]-4-yl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**3p**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3p** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (79.1 mg, 81%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.58 (d, J = 7.6 Hz, 2H), 7.48 (d, J = 7.6 Hz, 2H), 7.43 (d, J = 7.6 Hz, 2H), 7.38 (d, J = 7.6 Hz, 2H), 7.29 (d, J = 8 Hz, 2H), 7.15 (d, J = 8.8 Hz, 3H), 7.09 (d, J = 15.2 Hz, 2H), 6.86 (d, J = 8 Hz, 2H), 4.58-4.51 (m, 1H), 3.33 (d, J = 7.2 Hz, 2H), 2.45 (s, 3H), 2.33 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 143.8, 140.2, 139.4, 139.0, 137.8, 136.0, 133.9, 133.2, 130.2, 130.1, 129.9, 129.6, 128.9, 128.8, 128.2, 127.5, 126.9,

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110.5, 35.7, 21.7, 21.2. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{29}H_{26}ClNO_2SNa$, 510.1270; found 510.1273.

*(E)-N-(3-(2,6-Dichloropyridin-3-yl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**3q**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3q** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (45.0 mg, 52%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.53 (d, J = 7.6 Hz, 2H), 7.32-7.27 (m, 5H), 7.18-7.12 (m, 2H), 7.08 (d, J = 7.2 Hz, 1H), 7.01 (d, J = 14 Hz, 1H), 6.90 (d, J = 8 Hz, 2H), 4.53−4.46 (m, 1H), 3.39 (d, J = 6.8 Hz, 2H), 2.44 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 144.1, 137.9, 135.4, 135.2, 134.9, 133.9, 131.4, 130.4, 130.1, 129.8, 129.7, 129.5, 127.7, 127.5, 126.9, 109.5, 33.8, 21.7. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{22}H_{19}Cl_2NO_2SNa$, 454.0411; found 454.0415.

*(E)-N-(3-(2-Chlorophenyl)prop-1-en-1-yl)-4-methyl-N-(4-nitrophenyl)benzenesulfonamide (**3r**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3r** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (48.7 mg, 55%). ^1H NMR (400 MHz, $CDCl_3$): δ = 8.19 (d, J = 8.8 Hz, 2H), 7.53 (d, J = 7.6 Hz, 2H), 7.33−7.29 (m, 3H), 7.21−7.14 (m, 4H), 7.09 (d, J = 8 Hz, 1H), 6.95 (d, J = 14 Hz, 1H), 4.66−4.59 (m, 1H), 3.42 (d, J = 6.8 Hz, 2H), 2.45 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 147.3, 144.6, 143.2, 137.4, 135.0, 133.9, 130.7, 130.2, 129.9, 129.6, 127.9, 127.4, 127.0, 124.8, 112.0, 33.9, 21.7, 14.2. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{22}H_{20}ClN_2O_4S$, 443.0832; found 443.0833.

*(E)-N-(3-(2-Chlorophenyl)prop-1-en-1-yl)-N-(4-methoxyphenyl)-4-methylbenzenesulfonamide (**3s**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3s** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (55.6 mg, 65%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.54 (d, J = 7.6 Hz, 2H), 7.31 (d, J = 7.6 Hz, 1H), 7.27 (d, J = 7.2 Hz, 2H), 7.17−7.12 (m, 2H), 7.08 (d, J = 9.6 Hz, 1H), 7.05 (d, J = 14.4 Hz, 1H), 6.84 (dd, J = 16.8, 8.8 Hz, 4H), 4.52−4.45 (m, 1H), 3.79 (s, 3H), 3.39 (d, J = 6.8 Hz, 2H), 2.44 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 159.7, 143.8, 138.2, 135.8, 133.9, 131.2, 130.9, 130.1, 129.6, 129.4, 128.9, 127.6, 127.5, 126.8, 114.6, 108.6, 55.4, 33.7, 21.6. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{23}H_{22}ClNO_3SNa$, 450.0907; found 450.0910.

*(E)-N-(3-(2-Chlorophenyl)prop-1-en-1-yl)-N-phenylmethanesulfonamide (**3t**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3t** by column chromatography (PE/EA = 10:1) on silica gel as a yellow oil (29.6 mg, 46%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.48−7.39 (m, 3H), 7.33−7.30 (m, 3H), 7.19−7.11 (m, 3H), 6.91 (d, J = 13.6 Hz, 1H), 4.66−4.59 (m, 1H), 3.41 (d, J = 7.2 Hz, 2H), 3.0 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 138.0, 136.6, 133.9, 130.1, 130.0, 129.9, 129.8, 129.5, 129.2, 127.7, 126.9, 109.1, 38.9, 33.7. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{16}H_{17}ClNO_2S$, 322.0669; found 322.0674.

*(E)-3-(2-Chlorophenyl)prop-1-en-1-yl-N-cinnamyl-4-methylbenzenesulfonamide (**3u**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **3u** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (30.7 mg, 35%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.67 (d, J = 7.6 Hz, 2H), 7.31−7.28 (m, 6H), 7.21 (d, J = 7.6 Hz, 2H), 7.12 (d, J = 7.2 Hz, 2H), 7.05 (d, J = 7.2 Hz, 1H), 6.75 (d, J = 14 Hz, 1H), 6.43 (d, J = 16 Hz, 1H), 5.98−5.93 (m, 1H), 5.04-4.97 (m, 1H), 4.15 (d, J = 4.8 Hz, 2H), 3.44 (d, J = 6.8 Hz, 2H), 2.40 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 143.8, 138.2, 136.2, 136.1, 133.9, 133.0, 129.8, 129.4, 128.5, 127.8, 127.6, 127.1, 126.8, 126.4, 123.0, 109.5, 47.8, 34.1, 21.6. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{25}H_{24}ClNO_2SNa$, 460.1114; found 460.1109.

General Procedures for Products 4. Allenamide (0.20 mmol), phenylboric acid (0.24 mmol), $Pd(OAc)_2$ (5.0 mg, 0.02 mmol), dppf (11.0 mg, 0.02 mmol), and K_3PO_4 (85.0 mg, 0. 40 mmol) were added to a reaction tube, Then, 1,4-dioxane (2 mL) was added under nitrogen atmosphere. The resulting mixture was stirred at 50 °C for 0.5 h (the heat source is an oil bath). After cooling the reaction mixture at room temperature, the aqueous layer was extracted with EtOAc (3 × 10 mL). The combined organic phase was dried over Na_2SO_4, and then concentrated in vacuo. The mixture was purified by silica gel column chromatography to give the products.

*(E)-N-(2-(2-Chlorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**4a**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4a** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (66.7 mg, 81%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.39 (d, J = 7.2 Hz, 2H), 7.24 (d, J = 4.4, 1H), 7.12 (s, 5H), 7.02 (m, 4H), 6.23 (s, 1H), 2.30 (s, 3H), 2.26 (s, 3H), 1.64 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 143.7, 140.7, 138.2, 137.0, 134.6, 133.2, 133.1, 130.2, 129.7, 129.6, 129.4, 128.6, 127.8, 127.6, 127.4, 126.7, 21.6, 21.1, 17.8.

HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{23}H_{23}ClNO_2S$, 412.1138; found 412.1140.

*(E)-N-(2-(4-Chlorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**4b**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4b** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (56.0 mg, 68%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.49 (d, J = 7.6 Hz, 2H), 7.29-7.24 (m, 6H), 7.10 (d, J = 7.6 Hz, 2H), 7.03 (d, J = 8 Hz, 2H), 6.48 (s, 1H), 2.42 (s, 3H), 2.33 (s, 3H), 1.85 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 143.9, 139.1, 138.5, 137.1, 134.6, 134.4, 133.4, 129.6, 129.5, 128.5, 127.8, 127.4, 127.2, 125.2, 21.6, 21.1, 16.1 HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{23}H_{23}ClNO_2S$, 412.1138; found 412.1140.

*(E)-N-(2-(2-Fluorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**4c**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4c** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (66.9 mg, 93%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.49 (d, J = 7.6 Hz, 2H), 7.25-7.22 (m, 4H), 7.11−7.09 (m, 4H), 7.05−7.00 6.4 (m, 2H), 6.48 (s, 1H), 2.40 (s, 3H), 2.34 (s, 3H), 1.81 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 160.1 (d, J = 246.3 Hz), 143.8, 138.5, 137.0, 134.4, 130.3 (d, J = 1.1 Hz), 130.0 (d, J = 33.1 Hz), 129.6, 129.5, 129.0 (d, J = 8.5 Hz), 127.9, 127.8 (d, J = 3.9 Hz), 127.3, 124.1 (d, J = 3.4 Hz), 115.9 (d, J = 22.6 Hz), 21.6, 21.1, 17.1 (d, J = 2.8 Hz). ^{19}F NMR (376 MHz, $CDCl_3$): δ = −114.7. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{23}H_{22}FNO_2SNa$, 418.1253; found 418.1256.

*(E)-N-(2-(2-Fluorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**4d**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4d** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (56.8 mg, 79%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.49 (d, J = 7.6 Hz, 2H), 7.30-7.24 (m, 3H), 7.10 (d, J = 6 Hz, 3H), 7.04 (d, J = 9.2 Hz, 3H), 6.99 (d, J = 8.4 Hz, 1H), 6.42 (s, 1H), 2.42 (s, 3H), 2.33 (s, 3H), 1.88 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 162.4 (d, J = 245.3 Hz), 143.8, 138.6, 137.0, 136.7 (d, J = 3.1 Hz), 135.4, 134.5, 129.6, 129.5, 127.8, 127.7 (d, J = 8 Hz), 127.1, 124.7 (d, J = 1 Hz), 115.2 (d, J = 21.2 Hz), 21.6, 21.1, 16.4. ^{19}F NMR (376 MHz, $CDCl_3$): δ = −114.6. HRMS (ESI/OTOF-Q) m/z: $[M + Na]^+$ calcd for $C_{23}H_{22}FNO_2SNa$, 418.1253; found 418.1256.

*(E)-N-(2-(2,4-Difluorophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**4e**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4e** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (64.5 mg, 78%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.48 (d, J = 8 Hz, 2H), 7.24 (d, J = 7.6 Hz, 2H), 7.21−7.17 (m, 1H), 7.10 (d, J = 7.6 Hz, 2H), 7.05 (d, J = 8 Hz, 2H), 6.85−6.75 (m, 2H), 6.44 (s, 1H), 2.41 (s, 3H), 2.34 (s, 3H), 1.78 (s, 3H). $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ = 162.1 (dd, J = 235.5, 11.9 Hz), 160.1 (dd, J = 237.4, 11.5 Hz), 143.9, 138.3, 137.1, 134.4, 130.4 (dd, J = 9.4, 5.4 Hz), 129.7, 129.5, 129.4 (d, J = 1.3 Hz), 127.9, 127.3, 125.2 (dd, J = 13.9, 3.9 Hz), 111.2 (dd, J = 20.9, 3.7 Hz), 104.2 (dd, J = 26.3, 25.5 Hz), 21.6, 21.1, 17.1 (d, J = 2.6 Hz).

^{19}F NMR (376 MHz, $CDCl_3$): δ = −110.5, 111.2. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{23}H_{22}F_2NO_2S$: 414.1339, found 414.1344.

*(E)-N-(2-(2-Bromophenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**4f**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4f** by column

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chromatography (PE/EA = 10:1) on silica gel as a yellow oil (69.4 mg, 76%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.48 (d, J = 7.6 Hz, 2H), 7.26 (d, J = 6.8 Hz, 2H), 7.21 (d, J = 7.6 Hz, 3H), 7.11 (s, 5H), 6.29 (s, 1H), 2.39 (s, 3H), 2.34 (s, 3H), 1.71 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 143.8, 142.8, 138.1, 137.0, 134.6, 134.4, 132.8, 130.3, 129.6, 129.4, 128.8, 127.8, 127.5, 127.3, 122.9, 21.6, 21.1, 18.0.

HRMS (ESI-TOF) m/z: [M − H]$^+$ calcd for $C_{23}H_{21}NO_2SBr$, 454.0476; found 478.0478.

*(E)-N-(2-(2-Methoxyphenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**4g**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4g** by column chromatography (PE/EA = 10:1) on silica gel as a yellow oil (46.5 mg, 57%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.49 (d, J = 7.6 Hz, 2H), 7.25−7.20 (m, 4H), 7.12−7.09 (m, 4H), 6.84 (d, J = 8 Hz, 2H), 6.32 (s, 1H), 3.75 (s, 3H), 2.38 (s, 3H), 2.33 (s, 3H), 1.80 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 157.2, 143.6, 138.9, 136.7, 135.1, 134.7, 130.7, 129.5, 129.3, 128.8, 127.9, 127.5, 127.1, 126.2, 120.5, 111.1, 55.5, 21.6, 21.1, 17.3. HRMS (ESI-TOF) m/z: [M + Na]$^+$ calcd for $C_{24}H_{25}NO_3NaS$, 430.1453; found 430.1453.

*(E)-N-(2-(4-Methoxyphenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**4h**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4h** by column chromatography (PE/EA = 10:1) on silica gel as a yellow oil (68.5 mg, 84%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.51 (d, J = 7.6 Hz, 2H), 7.28−7.24 (m, 4H), 7.09 (d, J = 8 Hz, 2H), 7.05 (d, J = 8.4 Hz, 2H), 6.85 (d, J = 8.4 Hz, 2H), 6.38 (s, 1H), 3.80 (s, 3H), 2.42 (s, 3H), 2.33 (s, 3H), 1.91 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 159.4, 143.7, 139.0, 136.9, 136.8, 134.6, 132.9, 129.6, 129.4, 127.9, 127.3, 127.0, 123.3, 113.7, 55.3, 21.6, 21.1, 16.2.

HRMS (ESI-TOF) m/z: [M + Na]$^+$ calcd for $C_{24}H_{25}NO_3NaS$, 430.1453; found 430.1453.

*(E)-N-(2-(4-(Tert-butyl)phenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**4i**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4i** by column chromatography (PE/EA = 15:1) on silica gel as a yellow oil (77.2 mg, 89%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.50 (d, J = 8 Hz, 2H), 7.34 (d, J = 8 Hz, 2H), 7.29−7.23 (m, 4H), 7.06(dd, J = 13.6, 8.4 Hz, 4H), 6.48 (s, 1H), 2.40 (s, 3H), 2.32 (s, 3H), 1.88 (s, 3H), 1.31 (s, 9H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 150.8, 143.8, 138.9, 137.7, 136.9, 136.1, 134.6, 129.6, 129.5, 127.9, 127.1, 125.8, 125.3, 124.3, 34.6, 31.3, 21.6, 21.1, 16.2 HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{27}H_{32}NO_2S$: 434.2154, found 434.2158.

*(E)-2-(1-(4-Methyl-N-(p-tolyl)phenylsulfonamido)prop-1-en-2-yl)phenyl 4-methylbenzenesulfonate (**4j**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4j** by column chromatography (PE/EA = 10:1) on silica gel as a yellow oil (61.3 mg, 56%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.54−7.48 (m, 4H), 7.23 (d, J = 6.8 Hz, 2H), 7.18−7.12 (m, 6H), 7.06 (d, J = 7.6 Hz, 2H), 6.32 (s, 1H), 2.40 (s, 6H), 2.36 (s, 3H), 1.47 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 146.6, 145.2, 143.8, 138.2, 137.2, 135.6, 134.3, 132.5, 130.6, 130.1, 129.6, 129.5, 129.4, 128.6, 128.4, 128.3, 127.9, 127.7, 127.0, 122.8, 21.7, 21.2, 17.1. HRMS (ESI/OTOF-Q) m/z: [M + Na]$^+$ calcd for $C_{30}H_{29}NO_5S_2Na$, 570.1385; found 570.1382.

*(E)-Methyl4-(1-(4-methyl-N-(p-tolyl)phenylsulfonamido)prop-1-en-2-yl)benzoate (**4k**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4k** by column chromatography (PE/EA = 5:1) on silica gel as a yellow oil (47.0 mg, 54%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.98 (d, J = 8 Hz, 2H), 7.49 (d, J = 7.6 Hz, 2H), 7.38 (d, J = 8 Hz, 2H), 7.26 (d, J = 7.2 Hz, 2H), 7.10 (d, J = 7.6 Hz, 2H), 7.04 (d, J = 8 Hz, 2H), 6.64 (s, 1H), 3.91 (s, 3H), 2.42 (s, 3H), 2.34 (s, 3H), 1.84 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 166.8, 145.4, 144.0, 138.3, 137.3, 134.4, 133.4, 129.7, 130.0, 129.0, 127.9, 127.4, 126.7, 126.0, 52.2, 21.6, 21.1, 16.0. HRMS (ESI/OTOF-Q) m/z: [M + Na]$^+$ calcd for $C_{25}H_{25}NO_4SNa$, 458.1402, found 458.1408.

*(E)-N-(2-(3-Acetylphenyl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)-benzenesulfonamide (**4l**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4l** by column chromatography (PE/EA = 5:1) on silica gel as a yellow oil (68.0 mg, 81%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.88−7.84 (m, 2H), 7.54 (d, J = 7.6 Hz, 1H), 7.50 (d, J = 7.6 Hz, 2H), 7.44−7.40 (m, 1H), 7.26 (d, J = 7.2 Hz, 2H), 7.11 (d, J = 8 Hz, 2H), 7.05 (J = 7.6 Hz, 2H), 6.55(s, 1H), 2.61 (s, 3H), 2.43 (s, 3H), 2.34 (s, 3H), 1.90 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 198.1, 143.9, 141.4, 138.4, 137.2, 134.7, 134.4, 130.8, 129.7, 129.5, 128.7, 127.9, 127.7, 127.3, 125.9, 125.7, 26.8, 21.6, 21.1, 16.3. HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{25}H_{26}NO_3S$: 420.1633, found 420.1639.

*(E)-N-(2-(2,6-Dichloropyridin-3-yl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**4m**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4m** by column chromatography (PE/EA = 10:1) on silica gel as a yellow oil (45.6 mg, 51%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.54 (d, J = 7.6 Hz, 1H), 7.44 (d, J = 7.2 Hz, 2H), 7.27−7.22 (m, 3H), 7.12 (d, J = 7.6 Hz, 2H), 7.06 (d, J = 7.6 Hz, 2H), 6.43 (s, 1H), 2.41 (s, 3H), 2.36 (s, 3H), 1.63 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 146.6, 145.2, 143.8, 138.2, 137.2141.0, 137.5, 136.2, 134.2, 129.7, 129.6, 129.4, 127.8, 127.7, 127.1, 123.0, 21.6, 21.1, 17.3.

HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{22}H_{21}Cl_2N_2O_2S$, 447.0701; found 447.0705.

*(E)-4-Methyl-N-(2-(naphthalen-2-yl)prop-1-en-1-yl)-N-(p-tolyl)-benzenesulfonamide (**4n**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4n** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (41.0 mg, 48%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.92−7.90 (m, 1H), 7.86−7.84 (m, 1H), 7.77 (d, J = 8 Hz, 1H), 7.50-7.46 (m, 4 H), 7.42-7.38 (m, 1H), 7.23 (d, J = 7.2 Hz, 1H), 7.19-7.11 (m, 6H), 6.28 (s, 1H), 2.37 (d, J = 3.6 Hz, 6H), 2.02 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 143.7, 139.7, 138.7, 137.6, 137.1, 134.4, 133.7, 131.5, 129.7, 129.4, 128.3, 127.9, 127.8, 127.2, 126.9, 125.9, 125.7, 125.3, 125.2, 21.6, 21.2, 19.3. HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{27}H_{26}NO_2S$, 428.1684; found 428.1685.

*(E)-N-(2-(4′-Ethyl-[1,1′-biphenyl]-4-yl)prop-1-en-1-yl)-4-methyl-N-(p-tolyl)benzenesulfonamide (**4o**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **40** by column chromatography (PE/EA = 15:1) on silica gel as a yellow solid (81.9 mg, 85%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.55−7.50 (m, 6H), 7.39 (d, J = 8 Hz, 2H), 7.28-7.25 (m, 4H), 7.08 (dd, J = 15.6, 8.4 Hz, 4H), 6.55 (s, 1H), 2.69 (dd, J = 14.8, 7.2 Hz, 2H), 2.42 (s, 3H), 2.34 (s, 3H), 1.92 (s, 3H), 1.28 (t, J = 7.6 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 143.7, 143.6, 140.5, 139.2, 138.7, 137.9, 136.9, 135.6, 134.5, 129.6, 129.4, 128.3, 127.9, 127.1, 126.9, 126.4, 124.7, 28.5, 21.6, 21.1, 16.1, 15.6. HRMS (ESI-TOF) m/z: [M + H]$^+$ calcd for $C_{31}H_{32}NO_2S$, 482.2154; found 482.2157.

*(E)-N-(4-Chlorophenyl)-N-(2-(2-chlorophenyl)prop-1-en-1-yl)-4-methylbenzenesulfonamide (**4p**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4p** by column chromatography (PE/EA = 10:1) on silica gel as a yellow oil (76.9 mg, 89%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.47 (d, J = 8 Hz, 2H), 7.34 (d, J = 5.6, 1H), 7.29(d, J = 8 Hz, 2H), 7.24-7.18 (m, 7H), 6.28 (s, 1H), 2.40 (s, 3H), 1.73 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 144.2, 140.3, 139.6, 134.4, 134.2, 133.0, 132.6, 130.1, 129.7, 129.6, 129.2, 128.8, 128.5, 127.7, 127.1, 126.8, 21.6, 17.9. HRMS (ESI/OTOF-Q) m/z: [M + Na]$^+$ calcd for $C_{22}H_{19}Cl_2NO_2S$ Na, 454.0411; found 454.0415.

*(E)-N-(2-(2-Chlorophenyl)prop-1-en-1-yl)-N-(4-methoxyphenyl)-4-methylbenzenesulfonamide (**4q**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4q** by column chromatography (PE/EA = 10:1) on silica gel as a yellow oil (79.6 mg, 93%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.46 (d, J = 8 Hz, 2H), 7.33 (d, J = 7.2, 1H), 7.22 (s, 5H), 7.12 (d, J = 8.4 Hz, 2H), 6.83 (d, J = 8 Hz, 2H), 6.32 (s, 1H), 3.82 (s, 3H), 2.40 (s, 3H), 1.72 (s, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 158.5, 143.8, 140.8, 134.4, 133.5, 133.1, 132.2, 130.2, 129.7, 129.4, 129.1, 128.6, 127.9, 127.8, 126.7, 114.1, 55.4, 21.6, 17.8. HRMS (ESI/OTOF-Q) m/z: [M + Na]$^+$ calcd for $C_{23}H_{22}ClNO_3S$ Na, 450.0907; found 450.0910.

*(E)-N-(2-(2-Chlorophenyl)prop-1-en-1-yl)-4-methyl-N-octylbenzenesulfonamide (**4v**).* For 0.2 mmol scale, the standard procedure of method was followed to provide **4v** by column chromatography (PE/

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EA = 15:1) on silica gel as a yellow oil (79.4 mg, 81%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.70 (d, J = 7.6 Hz, 2H), 7.38−7.36 (m, 1H), 7.29 (d, J = 8.0 Hz, 2H), 7.26−7.18 (m, 3H), 5.14 (s, 1H), 3.08 (t, J = 6.8 Hz, 2H), 2.40 (s, 3H), 2.15 (s, 3H), 1.59−1.56 (m, 2H), 1.27−1.24 (m, 10H), 0.88 (t, J = 5.2 Hz, 3H). ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$): δ = 143.4, 143.3, 140.5, 135.0, 132.5, 129.9, 129.6, 129.5, 128.8, 127.6, 126.7, 126.0, 50.7, 31.8, 29.2, 29.1, 28.2, 26.7, 22.6, 21.5, 18.5.

Larger-Scale Synthesis. To further explore the potential of this methodology, a larger-scale operation was practically performed on 1 mmol for the synthesis of (E)-*N*-(3-(2-chlorophenyl)prop-1-en-1-yl)-4-methyl-*N*-(*p*-tolyl)benzenesulfonamide **3a** with a significant synthetic tool value in an 82% isolated yield (337.8 mg).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.1c01303.

Experimental procedures, optimization studies, characterization data, and NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

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